

**BLM4800- Introduction to Data Mining Syllabus**

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Course Hours Fri 14:00--16:50PM
Office Hours Fri 13:00--14:00PM or by appointment

Text Books

- Data Mining , J. Han – M. Kamber, Morgan-Kaufman, Academic Press, 2001, ISBN: 1-55860-901-6
- Data Mining – Concepts, Models, Methods and Algorithms, Mehmed Kantardzic, ISBN:0-471-22852-4

Course Description

Data mining is discovery of the knowledge from the huge amount of data. The objective of data mining is to find useful patterns in bulky data and to use discovered patterns to help explain behavior or to predict future outcomes. Data Mining Concepts, Preparing the Data, Statistical Classification Method (Naïve Bayes), Clustering Methods(K-Means,Hierarchical), Decision Trees and Decision Rules, Association Rules.

Upon completion of the course, students will:

- 1- have the basic knowledge and application ability about the data mining methods.
- 2- have the ability of exploring the knowledge from bulky data.
- 3- have the ability to analyze, clear and find the outliers from bulky data.
- 4- have knowledge about supervised classification methods.
- 5- have knowledge about unsupervised clustering methods.

Course Format

The lectures will include readings tutorials. The PowerPoint presentations/class notes will also be available on the website following each class. During class meetings, there will be some lecturing. You will tackle different data mining problems individually.

Participation: Participation will be a part of your course grade. You will be responsible for recovering any information you have missed if you do not attend a particular day's lecture.

Exams: The exams will be 90-120 minutes, closed book, closed notes exam. The use of any reference material is strictly forbidden.

Project: There will be one project for the course. It can be a group project or an individual project. For group project, the students are expected to provide more comprehensive work. For group project, all study group members will receive the same score. In general, students may freely communicate within their group, but you may not discuss group work with members of other groups. Details of this assignment will be discussed later in the semester.

Students are responsible for forming and managing their study groups. We expect that students will manage their study groups so that everyone performs a fair share of the work, and that all perspectives are heard and considered. The assignments should be turned in electronically by end of the due date.

Correspondence: If you want to get response to your e-mails, always include your name, your course name. Observe grammatical rules while composing your e-mails.

Grading

Final course grades will be based on:

Project	30%
Midterm Examination	30%
Final Examination	40%

No late assignments will be accepted. No make-up will be administered for the midterm.

TENTATIVE SCHEDULE

WEEK 1	March /3
Topics to be Covered:	Introduction to Data Mining
WEEK 2	March /10
Topics to be Covered:	Data Mining Concepts
WEEK 3	March /17
Topics to be Covered:	Preparing the Data
WEEK 4	March /24
Topics to be Covered:	Data Reduction
WEEK 5	March /31
Topics to be Covered:	Statistical Classification Method (Naïve Bayes)
WEEK 6	April /7
Topics to be Covered:	Decision Trees and Decision Rules
WEEK 7	April /14
Topics to be Covered:	Clustering and Similarity Measures
WEEK 8	April /28
Topics to be Covered:	MIDTERM EXAM
WEEK 9	May /5
Topics to be Covered:	Clustering Methods(Hierarchical Clustering)
WEEK 10	May /12
Topics to be Covered:	Evaluation of Classification Methods
WEEK 11	May /26
Topics to be Covered:	Association Rule Mining

Covered:	
WEEK 12	June /2
Topics to be Covered:	Classification with Artificial Neural Networks
WEEK 13	June /3
Topics to be Covered:	PRESENTATIONS
WEEK 14	June /3
Topics to be Covered:	PRESENTATIONS
WEEK 15	
Topics to be Covered:	FINAL